MILITARY SPECIFICATION
TEST SET,
PNEUMATIC PRESSURE
AN/PSM-15C

This Specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for flight line and shop use. The test equipment shall provide adjustable and regulated, self-generated, pneumatic pressures for simulating altitudes and airspeeds during ground testing of aircraft air data sensing equipment. The test equipment shall further provide regulation of pressure and vacuum from an external source to simulate engine bleed air to test the Air Data Computer Set, CPK-30/A24G-12 and the Angle of Attack Transmitter, TRK-48/A24G-12.

1.2 Classification - The TEST SET, PNEUMATIC PRESSURE AN/PSM-15C shall consist of the following items:

ITEM	TYPE DESIGNATION	APPL. PARA.
Simulator, Pneumatic	SM-392(*)/PSM-15C	3.4.1
Cover, Test Set	CW-756(*)/PSM-15C	3.4.2
Cable Assembly, Power	W1 of AN/PSM-15C	3.4.3.1
Cable Assembly, Ground	W2 of AN/PSM-15C	3.4.3.2
Hose Assembly, Pneumatic, Pitot Pressure	-2 -3	3.4.4.1
Hose Assembly, Pneumatic, Static Pressure (Corrected)	-4	3.4.4.2
Hose Assembly, Pneumatic, Engine Bleed	-5	3.4.4.3
Hose Assembly, Pneumatic, Static Pressure (Indicated)	-6	3.4.4.4
Wrench, Hexagon	0.049 Flats	3.4.5.1
Wrench, Hexagon	0.187 Flats	3.4.5.2
Wrench, Hexagon	0.0615 Flats	3.4.5.3

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionics Equipment, General Specification for

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS-21993	Cable Assemblies, Power, Electrical

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2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used where practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 84 pounds.

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3.2.2 Reliability

3.2.2.1 Operational Stability - The equipment shall operate with satisfactory performance continuously or intermittently for a period of at least 50 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and, the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.2.2.3 Reliability in Mean-Time-Between-Failures (MTBF) - The equipment shall have a minimum of 500 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.2.4 Time Totalizing Meter - The equipment shall incorporate a time totalizing meter in accordance with MIL-M-7793. The meter shall provide a digital readout of total ON time up to 9999 hours and shall be non-resettable.

3.2.3 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200. Insofar as is practicable, all connectors on the front panel shall be mounted along the upper edge of the panel and in no case shall their location be such that the connecting cables will interfere with the operating controls.

3.2.3.1 Cable Assemblies, Power, Electrical - Applicable input power cable and connections shall conform to MS-21993.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.5 Interference Control - The generation of radio interference and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

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3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient
Input Power Voltage	115 VAC ±10%, 3-phase 400 ± 20 cps

3.2.9 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 2 equipment, except as modified herein.

3.2.9.1 Vibration - The equipment shall operate satisfactorily when subjected to vibration.

3.2.10 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed one minute under standard conditions or 10 minutes at extreme service conditions.

3.2.11 Primary Input Power Requirements - The test equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in Specification MIL-T-21200 for Class 2 equipment. The power required shall be of the following type and shall not exceed the specified amount.

AC Power, Three Phase, 4 wire, neutral ground, wye connection, 115V, 400 cps, 760 VA maximum

3.2.11.1 Voltage Protection - The equipment shall not be damaged when exposed to transient voltages as defined in Figure 2 of MIL-STD-704, and shall resume normal operation when the voltage returns to within limits of normal operation and when the Power switch is operated to OFF and returned to ON.

3.3 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

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3.3.1 Static Pressure - The AN/PSM-15C shall provide static pressures which simulate conditions at altitudes between -1500 and 80,000 feet, with an accuracy of 0.5 percent or 30 feet, whichever is greater, when correlated with the correction cards supplied in the test set (see Figure 1 herein). Under reduced performance conditions, the accuracy shall be within 1.0 percent or 100 feet, whichever is greater.

3.3.1.1 Rate of Change - The rate of change of static pressure shall be manually controllable, by means of an airflow regulator, between zero feet per minute and 35,000 feet per minute.

3.3.2 Impact Pressure - The AN/PSM-15C shall provide impact pressures simulating those encountered at airspeeds between 80 and 850 knots, with an accuracy of 1.176 percent or 30 knots, whichever is greater, when correlated with the correction cards supplied in the test set (see Figure 1 herein). Under reduced performance conditions, the accuracy shall be within 4.0 percent.

3.3.3 Mach Number - The AN/PSM-15C shall provide static and impact pressures that simulate those encountered at a speed of 2.5 Mach at altitudes between 36,000 and 80,000 feet, with an accuracy of +0.1, -0.0 Mach.

3.3.4 Engine Bleed Air - The AN/PSM-15C, when furnished with an air pressure supply of 250-3000 pounds per square inch, gage (psig), shall provide a regulated, simulated engine bleed air flow. This airflow shall be adjustable between 0 and 255 $\pm$ 5 psig and between 0 and 0.25 pound per minute flow.

3.3.5 Drift - The simulated static or impact pressures provided by the AN/PSM-15C shall not drift more than 0.004 inch of mercury during any five minute period, after stabilization.

3.3.6 Function - The AN/PSM-15C shall provide static and impact pressures to simulate those pneumatic pressures encountered by the aircraft as a result of speed and altitude and shall regulate an external airflow source simulating engine bleed air, for the purpose of performing checkout tests on air data sensing equipment used in aircraft.

3.3.7 Form Factor - The dimensions of the AN/PSM-15C, including the combination case and cover, shall not exceed 16.0 inches wide by 15 inches high by 21.5 inches long.

3.4 Detail Requirements

3.4.1 Simulator, Pneumatic SM-392/PSM-15C - The Pneumatic Simulator shall contain the following functional sections and items on the panel:

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3.4.1.1 Power Section - This section of the Pneumatic Simulator Panel shall be labeled POWER and shall contain the following items:

3.4.1.1.1 Power Receptacle - This receptacle, J1, shall provide for interconnection between the Pneumatic Simulator and the electrical power source via Cable Assembly W1.

3.4.1.1.2 Power On-Off Switch - This 2-position toggle switch shall have positions labeled ON (up) and OFF (down). The switch shall control application of electrical power to the Pneumatic Simulator.

3.4.1.1.3 Indicator Lamp - This red indicator lamp, labeled ON, shall light when the switch of 3.4.1.1.2 is in the ON position and power is present. The lamp shall continue to glow while the switch of 3.4.1.1.2 is in the ON position and power is applied to the circuitry.

3.4.1.1.4 System Ready Indicator - This amber indicator lamp, labeled SYSTEM READY, shall light within 15 seconds after the application of power and shall remain lit while the switch of 3.4.1.1.2 is in the ON position and power is applied to the circuitry.

3.4.1.2 Engine Bleed Air Section - This section of the Pneumatic Simulator Panel shall be labeled ENGINE BLEED and shall contain the following items:

3.4.1.2.1 Regulator Control - This knob, labeled REGULATOR, shall be further identified with a clockwise arrow labeled INCREASE. The control shall provide for regulation of the output of simulated engine bleed air.

3.4.1.2.2 Pressure Gauge - This dial indicating gauge, labeled OUTPUT PRESS, shall indicate pressure, in pounds per square inch, from 0 to 300. The position of the needle shall indicate output pressure from the regulator of 3.4.1.2.1.

3.4.1.2.3 Pneumatic Fitting, 250 to 3000 PSI Supply - This pneumatic fitting, labeled 250-3000 PSI SUPPLY, shall provide for connection of the supply pressure to the Engine Bleed Regulator.

3.4.1.2.4 Pneumatic Fitting, Regulated Output - This pneumatic fitting, labeled REGULATED OUTPUT, shall provide for connection between the Pneumatic Simulator and the Engine Bleed connection on the Static Pressure Compensator.

3.4.1.3 Pressure/Vacuum Source Section - This section of the Pneumatic Simulator Panel shall be labeled SOURCE and shall contain the following items

3.4.1.3.1 Pneumatic Fitting, Pressure - This pneumatic fitting, labeled PRESSURE, shall provide for connection between the Pneumatic Simulator Panel and the component under test.

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3.4.1.3.2 Pneumatic Fitting, Vacuum - This pneumatic fitting, label VACUUM, shall provide for connection between the Pneumatic Simulator Panel and the component under test.

3.4.1.4 Altitude Rate Control - This continuously variable pressure control (valve), labeled ALT RATE ADJUST, shall be further identified with a counterclockwise arrow labeled INCREASE. The control shall regulate the rate of change of simulated altitude.

3.4.1.5 Circuit Breaker - This 5-amp circuit breaker, labeled CIRCUIT BREAKER, shall be further idnetified with the label PUSH TO RESET. The circuit breaker shall provide protection to the Pneumatic Simulator against current overloads.

3.4.1.6 Pneumatic Fitting, Corrected Static Pressure - This pneumatic fitting, labeled CORRECTED STATIC PRESS., shall be further identified with the label FROM ACFT COMPENSATOR. The fitting shall provide for connection between the Pneumatic Simulator and the aircraft.

3.4.1.7 Pneumatic Fitting, Pitot Pressure - This penumatic fitting, labeled PITOT PRESS., shall share with the pneumatic fitting of 3.4.1.8 the label TO ACFT. The fitting shall provide for connection between the Pneumatic Simulator and the aircraft.

3.4.1.8 Pneumatic Fitting, Indicated Static Pressure - This pneumatic fitting, labeled INDICATED STATIC PRESS., shall share with the pneumatic fitting of 3.4.1.7 the label TO ACFT. The fitting shall provide for connection between the Pneumatic Simulator and the aircraft.

3.4.1.9 Vents - Two vents, unlabeled, shall be provided on the Pneumatic Simulator Panel. These vents shall provide the means for relieving internal pressure or vacuum.

3.4.1.10 Motor Power Section - This section of the Pneumatic Simulator Panel shall be labeled MOTOR POWER and shall contain the following items:

3.4.1.10.1 Indicator, Phase Unbalance - This lamp, labeled PHASE UNBALANCE, shall light when the Pneumatic Simulator is not receiving proper phase balance, thus indicating an input power unbalanced condition.

3.4.1.10.2 Ground Receptacle - This receptacle, J?, shall provide for interconnection between the Pneumatic Simulator and ground via Cable Assembly W2.

3.4.1.11 Altimeter Pressure Select Switch - This 2-position toggle switch, equipped with a dust cap, shall have positions labeled CORRECTED STATIC (up) and INDICATED STATIC ALTIMETER PRESS. SEL (down). The switch shall control the test set altimeter for monitoring the test set vacuum output when the switch is in the down position.

When the switch is in the up position, the test set altimeter shall monitor the corrected static pressure from an aircraft that has a static air correction system.

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3.4.1.12 Airspeed Leak Switch - This 2-position spring return toggle switch, equipped with a dust cap, shall have positions labeled AS. LEAK (up) and OFF (down). The switch shall share with the switch of 3.4.1.13 the label AIRCRAFT LEAK TEST. Placing the switch in the AS. LEAK position shall close the pressure line from the test set. Any leak in the component or system under test shall be monitored on the indicator of 3.4.1.14.

3.4.1.13 Altitude Leak Switch - This 2-position spring return toggle switch, equipped with a dust cap, shall have positions labeled ALT LEAK (up) and OFF (down). The switch shall share the label AIRCRAFT LEAK TEST with the switch of 3.4.1.12. Placing the switch in the ALT LEAK position shall close the vacuum line from the test set. Any leak in the component or system under test shall be monitored by the indicator of 3.4.1.15.

3.4.1.13.1 Rocker Arm - The switches of 3.4.1.12 and 3.4.1.13 shall be harnessed by a rocker arm which will permit both switches to be OFF at the same time and either switch to be ON while the other is OFF, but will not permit both switches to be ON at the same time.

3.4.1.4 Airspeed Indicator - This dial indicator, labeled AIR-SPEED, shall provide two functions, as follows:

(1) Monitor the test set pitot output (simulated airspeed output).

(2) When used in conjunction with the Airspeed Leak Switch of 3.4.2.12, monitor the leakage in the component or system under test.

3.4.1.14.1 Indicated Airpseed Control - This control, labeled INDICATED AIRSPEED, shall be further identified with a clockwise arrow labeled INCREASE. Turning the control shall provide for adjustment of the pitot output from the test set.

3.4.1.15 Altitude Indicator - This indicator (altimeter), labeled ALTITUDE, shall provide three functions, as follows:

(1) Monitor the test set static pressure (simulated altitude output).

(2) When used in conjunction with the Altitude Leak Switch of 3.4.2.13, monitor the leakage in the component or system under test.

(3) When used in conjunction with the Altimeter Pressure Select switch of 3.4.1.11, provide for comparison of the difference between indicated static pressure and corrected static pressure.

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3.4.1.15.1 Indicated Altitude Control - This control, labeled INDICATED ALTITUDE, shall be further identified with a counterclockwise arrow labeled INCREASE. Turning the control shall provide for adjustment of the static output from the test set.

3.4.1.16 Nameplate - A nameplate conforming to MIL-N-18307 shall be fastened to the Pneumatic Simulator Panel.

3.4.2 Cover, Test Set CW-756/PSM-15C - The combination case shall be provided with a cover conforming to the applicable requirements of Specification MIL-T-21200.

3.4.2.1 Function - The cover shall provide protection and storage for the cable assemblies, hose assemblies, and accessories of the AN/PSM-15C. Also, a schematic placard and an operating instructions placard shall be provided inside the cover.

3.4.2.2 Form Factor - The dimensions of the cover shall not exceed 15.0 inches wide by 7.25 inches high by 20.0 inches long.

3.4.2.3 Weight - The weight of the cover, including cable assemblies, hose assemblies, and accessories, shall not exceed 18 pounds.

3.4.2.4 Contents of Unit - The cover shall contain, in addition to the electrical schematic and operating instructions placards, one of each of the following items:

(a) Cable Assembly W1 of AN/PSM-15C

(b) Cable Assembly W2 of AN/PSM-15C

(c) Hose Assembly, Pneumatic, -3, (pitot pressure)

(d) Hose Assembly, Pneumatic, -4, (corrected static pressure)

(e) Hose Assembly, Pneumatic, -5, (engine bleed)

(f) Hose Assembly, Pneumatic, -6, (indicated static pressure)

(g) Wrench, Hexagon, 0.049 Flats

(h) Wrench, Hexagon, 0.187 Flats

(i) Wrench, Hexagon, 0.0615 Flats

3.4.3 Cable Assemblies

3.4.3.1 Cable Assembly, Power W1 of AN/PSM-15C

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3.4.3.1.1 Function - The W1 Cable Assembly shall provide the means for connecting three phase primary power from an external source to the test set (see 3.4.1.1.1).

3.4.3.1.2 Form Factor - The W1 Cable Assembly shall not exceed 210 inches in length and 0.365 inch in diameter. The W1 Cable Assembly shall be terminated at one end with a connector, designated as P1, which shall mate with connector J1 on the test set. The other end of the cable assembly shall be terminated with a connector, designated as P3, which shall connect to an external power source as specified in 3.2.11. The P1 connector shall conform to MS 3106A-14S-2S; connector P3 shall conform to MS 3107A-20-4P.

3.4.3.1.3 Weight - The weight of the W1 Cable Assembly shall not exceed two pounds.

3.4.3.1.4 Pin Assignments - Electrical connections shall be made using size 18 AWG copper wire. Pin assignments shall be as follows:

P1A/P3A	Phase A, 115 VAC, 400 CPS
P1B/P3B	Phase B, 115 VAC, 400 CPS
P1C/P3C	Phase C, 115 VAC, 400 CPS
P1D/P3D	Neutral

3.4.3.2 Cable Assembly, Ground W2 of AN/PSM-15C

3.4.3.2.1 Function - The W2 Cable Assembly shall provide the means for connecting the test set chassis to ground (earth) (see 3.4.1.10.2).

3.4.3.2.2 Form Factor - The W2 Cable Assembly shall not exceed 210 inches in length and 0.182 inch in diameter. The W2 Cable Assembly shall be terminated at one end with a connector, designated as P2, which shall mate with connector J2 on the test set. The other end of the cable assembly shall be terminated with an alligator clip, designated as E1, which shall connect to an external earth ground. The P2 connector shall conform to MS 3106A-8S-1S.

3.4.3.2.3 Weight - The weight of the W2 Cable Assembly shall not exceed one pound.

3.4.4 Hose Assemblies

3.4.4.1 Hose Assembly, Pneumatic, -3, Pitot Pressure - The -3 pneumatic hose assembly shall provide the means for connecting the pitot pressure output of the test set (see 3.4.1.7) to the pitot pressure system of the aircraft. The -3 hose assembly shall not exceed 180 inches in length and should be 0.405 inch in diameter.

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3.4.4.2 Hose Assembly, Pneumatic, -4, Corrected Static Pressure - The -4 pneumatic hose assembly shall provide the means for connecting the corrected static pressure output of the test set (see 3.4.1.6) to the Central Air Data Computer (CADC) of the aircraft. The -4 hose assembly shall not exceed 180 inches in length and should be 0.480 inch in diameter.

3.4.4.3 Hose Assembly, Pneumatic, -5, Engine Bleed - The -5 pneumatic hose assembly shall provide the means for connecting the engine bleed regulated output of the test set (see 3.4.1.2.4) to the CADC of the aircraft. The -5 hose assembly shall not exceed 180 inches in length and should be 0.585 inch in diameter.

3.4.4.4 Hose Assembly, Pneumatic, -6, Indicated Static Pressure - The -6 pneumatic hose assembly shall provide the means for connecting the indicated static pressure output of the test set (see 3.4.1.8) to the static pressure system of the aircraft. The -6 hose assembly shall not exceed 180 inches in length and should be 0.630 inch in diameter.

3.4.5 Wrenches

3.4.5.1 Wrench, Hexagon, 0.049 Flats - The 0.049 wrench shall be supplied to adjust the ball screw assembly of the limit stop mechanism used to maintain the specified 2.5 Mach limit.

3.4.5.2 Wrench, Hexagon, 0.187 Flats - The 0.187 wrench shall be supplied to adjust the stops of the limit stop mechanism used to maintain the specified 2.5 Mach limit.

3.4.5.3 Wrench, Hexagon, 0.0615 Flats - The 0.0615 wrench shall be supplied to adjust the coupler assembly of the limit stop mechanism to maintain the specified 2.5 Mach limit.

3.5 Associated Equipment - To operate the Engine Bleed Air section of the AN/PSM-15C (see 3.4.2.2) a source of air pressure/vacuum in the 250-3000 psi range is required. However, this air pressure/vacuum source is not supplied as a part of the AN/PSM-15C.

3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Service Approval Tests - At the completion of contractor demonstration tests, and when requested by the procuring activity, the equipment shall be delivered to specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

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4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental testing in accordance with Specification MIL-T-5422, including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent by the contractor at his expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is accomplished and approved. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

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4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

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4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production, and special tests.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be of two classifications:

a. Those conducted prior to delivery of any equipment under the contract or order

b. Those conducted on samples during the production run.

Unless otherwise stipulated in the contract, equipments which have successfully passed the individual tests shall be tested for reliability assurance. Test level II of Specification MIL-R-23094 shall be used. Detail test procedures for conducting all reliability assurance tests shall be prepared by the contractor and approved by the procuring activity. The detail test procedures must include (but not be limited to) the performance characteristics to be measured during the tests, details of the cycle, failure criteria provisions, preventive maintenance allowed and other special requirements. All equipments used shall be refurbished to "as-good-as-new" and delivered after tests:

4.4.3.1 Reliability Assurance Tests conducted Prior to Delivery - Prior to the delivery of any equipment the contractor must demonstrate the MTBF requirement of paragraph 3.2.2.3 by placing a minimum of three equipments on test. One of the three equipments must operate for at least 300 hours. Using Figure 15 of Specification MIL-R-23094, the MTBF will have been demonstrated when an accept decision is reached. One unit will remain on test after delivery begins until 1500 hours have been accumulated thereon and the results summarized as indicated below.

4.4.3.2 Reliability Assurance Tests Conducted on Samples - Using Procedure II of Specification MIL-R-23094, sample equipments shall be placed on test throughout the life of the contract. At least 150 hours of test time shall be logged per month. A summary of the results shall be forwarded to the procuring activity at the conclusion of each month.

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4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipment for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special TestsSchedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See paragraph 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Tests procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.5 Equipment Failure - Should a failure occur during testing, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

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(4) When practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of tests procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity. The test procedure for the reliability tests shall be prepared by the contractor and approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The AN/PSM-15C is intended for use on the flight line or in the shop to perform checkout tests on air data sensing equipment used in aircraft.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1.0 VAC, 400 ± 20 CPS, 3-phase, 4-wire, neutral ground. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

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6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Eachrequest shall be accompanied by complete supporting information.

6.5 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Design Data.

6.6 The parentheses (*), when used in the type designation, may be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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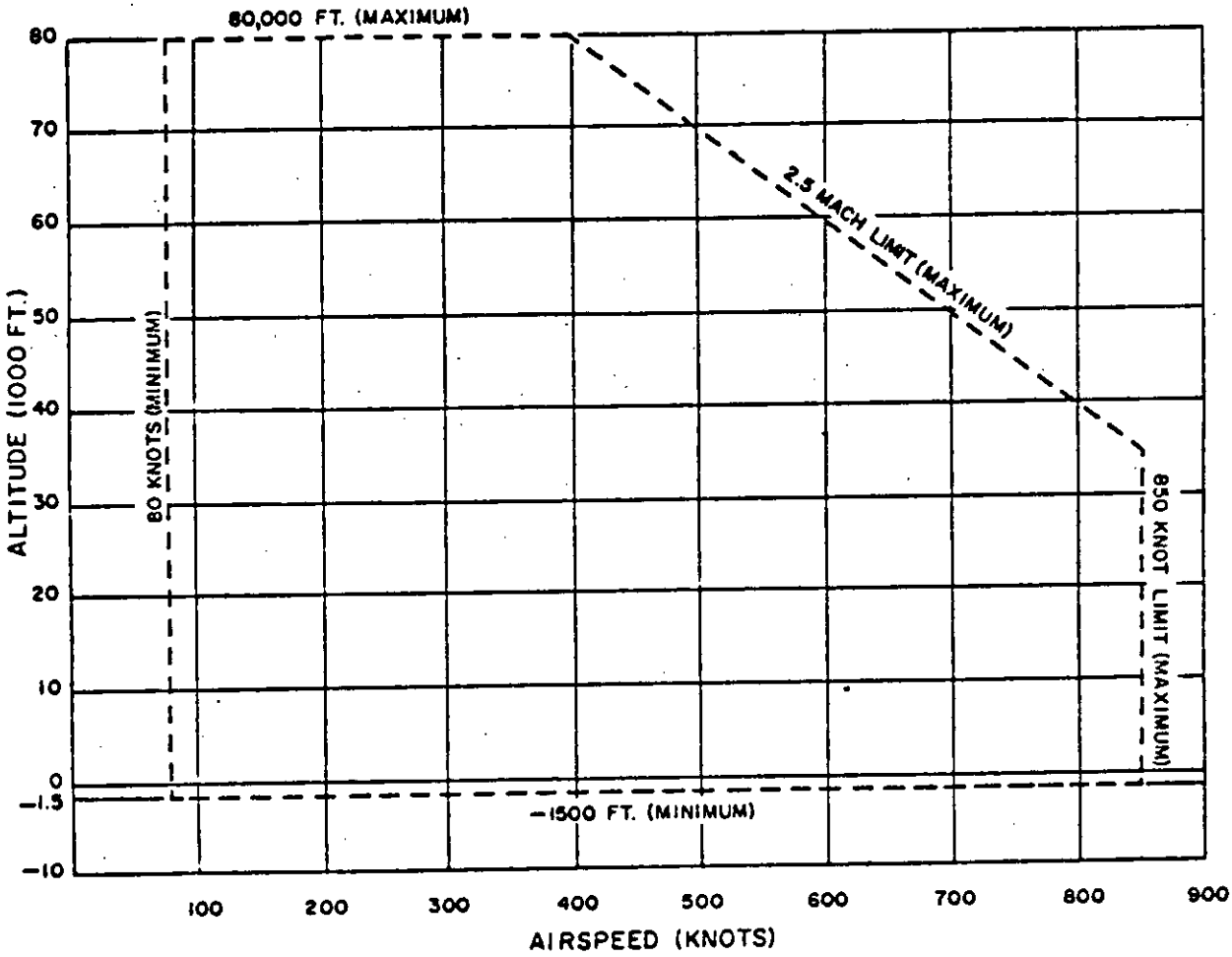


FIGURE 1

OPERATIONAL ENVELOPE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION
MIL-T-81223(WP) TEST SET PNEUMATIC PRESSURE AN/PSM-15C

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426

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